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Nimodipine and Microsurgery Induced Recovery of the Vocal Cord after Recurrent Laryngeal Nerve Resection

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Transection of the recurrent laryngeal nerve leads to permanent palsy of the vocal cord. Experimental studies have confirmed that nimodipine increases the pace of axonal regeneration. We present a case of a 19-year-old male, suffering a thyroid cancer disease, who was subjected to unilateral resection of the recurrent laryngeal nerve during surgery. The nerve was repaired with a nerve graft and the patient further treated with nimodipine for 3 months. Evaluation of the patient showed normalization of voice, movement of the vocal cord on the injured side, and electromyography evidence of reinnervation of the larynx muscles at 15 months after surgery. *Key Words:* Nimodipine, nerve graft, nerve injury, electromyography.

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INTRODUCTION

Up to the present, few options have been available for intraoperative damage to the recurrent laryngeal nerve (RLN). History reveals numerous attempts of direct anastomosis with poor results.[1,2] Treatment options have been focused on voice training, vocal cord injections or sometimes nerve anastomosis with other nerves than the RLN, the procedures of which are focused on trophy and medialization of the injured side as well as a demarked spasmic dysphonia.[3,4]

Experimental research has, over a century, exposed parameters important for regeneration and functional restitution after peripheral nerve injury.[5] They may be summarized by cortical contact, neuronal survival, pace of regeneration, misguidance, and degree of muscle atrophy. After injury to the RLN, there have been speculations about misguidance into the adductor and abductor branches of the nerve causing the loss of functional restitution, hampered coordination, or synkinesis of the vocal cord.[6] There is also evidence of rapid occupation of motor end plates in the larynx muscles by autonomous fibers after RLN injury.[7] Recent data suggest that neuronal survival after severe injury to the RLN is high and a minor problem.[6] The calcium flow antagonist nimodipine has been shown to speed up regeneration by a factor of two and also to improve functional restitution.[8] In light of earlier positive experiences in patients with nimodipine (unpublished) and the striking experimental findings on regeneration, we decided to combine microsurgery and medical treatment in a patient subjected to resection of the RLN during thyroid tumor surgery.

CASE

Surgery

A 19-year-old, South-American male with a locally advanced papillary thyroid cancer was subjected to surgery where the right RLN was embedded in the tumor and peroperatively resected by 30 mm at the level of the thyroid about 25 mm before entrance into the larynx muscles, proximal to the division into the adductor and abductor branches. A sural nerve graft was harvested from the right leg, and, to avoid anastomosis tension, 50 mm of the graft was used to cover the 30 mm diastasis of the proximal and distal ends of the RLN. Using 9–0 Prolene suture, three epineural sutures were placed at each anastomosis under the microscope. The nerve graft, including the anastomosis areas, were embedded in Surgicel (Ethicon, Johnsson & Johnsson, Stockholm, Sweden), which was folded around the nerve, and finally the graft area was covered with Ticeel (Baxter Inc., Stockholm, Sweden) to protect the anastomosis areas from tension when moving the head and to avoid blood components in the anastomosis.

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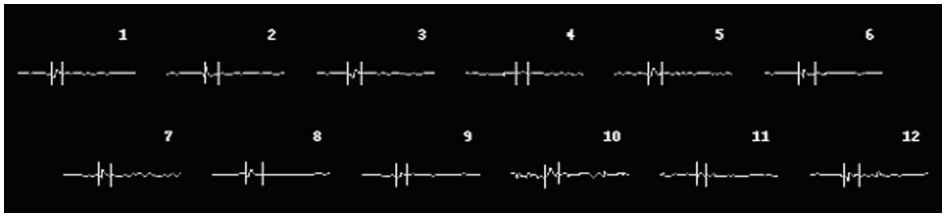


Fig. 1. Electromyography (EMG) from m. vocalis dx (injured side) showing a picture of total denervation with many fibrillation potentials and positive sharp waves and no motor unit potentials (MUPs) at 4 months postlesion.

Nimodipine Treatment

The patient was put on nimodipine infusion (1 mg/h), starting 10 hours postoperatively. He was intubated for 2 days because of extended surgery time and respiratory insufficiency, subsequently extubated, and put on oral nimodipine (60 mg ×3) for 3 months, including a successive decrease the last 2 weeks.

Electromyography

Quantitative concentric needle electromyography (EMG) was performed in the m. vocalis (m. thyroarytenoideus) bilaterally, repeatedly up to 15 months postoperatively, by a percutaneously inserted needle with the patient awake. At rest, denervation activity was searched for by detection of fibrillation potentials and positive sharp waves. In the study of voluntary activation, motor unit potentials (MUP) were gathered, and in maximal activation, the interference pattern was studied.

Laryngoscopy

To evaluate the motion and level differences of the vocal folds, video laryngostroboscopy was performed four times during the first 15 months postoperatively. We used a 70 degree rigid endoscope connected to DV-CAM recorder and stroboscopic equipment. The patient phonated on /i:/ vowel as close as possible to the habitual speaking pitch at a comfortable vocal effort. The ab-adduction of the vocal folds, the level difference between the folds, and the glottal closure were judged.

RESULTS

Electromyography

The patient was examined at 4 and 15 months postoperatively. The left, uninjured, RLN showed normal activity throughout the study period. At 4 months, ipsilateral to the nerve injury, EMG showed a picture of total denervation with many fibrillation potentials and positive sharp waves and no MUPs (Fig. 1).

At 15 months after nerve reconstruction, ipsilateral to the nerve injury, there was no denervation activity. The MUPs were found to be heterogeneous in amplitude and duration, representing small and large motor units. There were also some MUPs with polyphasic appearance, reflecting still ongoing reinnervation (Fig. 2). The collected MUPs indicated a very good reinnervation.

Laryngoscopy

The patient was examined at 3, 4, 7, and 13 months postoperatively. At 3 months, the voice was hoarse with voice breaks, just like the quality at dismissal from the hospital. There was then a gradual increase in voice quality, and, at the end of the study period, the voice was subjectively and objectively normal, including singing at high-pitched tones and loud shouting. The laryngoscopy correlated well with the voice quality, initially showing no movement of the ipsilateral vocal cord and later, small fibrillations. At the last examination, voluntary ab- and adduction of the vocal cord was seen together with complete glottic closure.

DISCUSSION

This is the first case to present strong evidence of reinnervation and functional recovery of the vocal cord

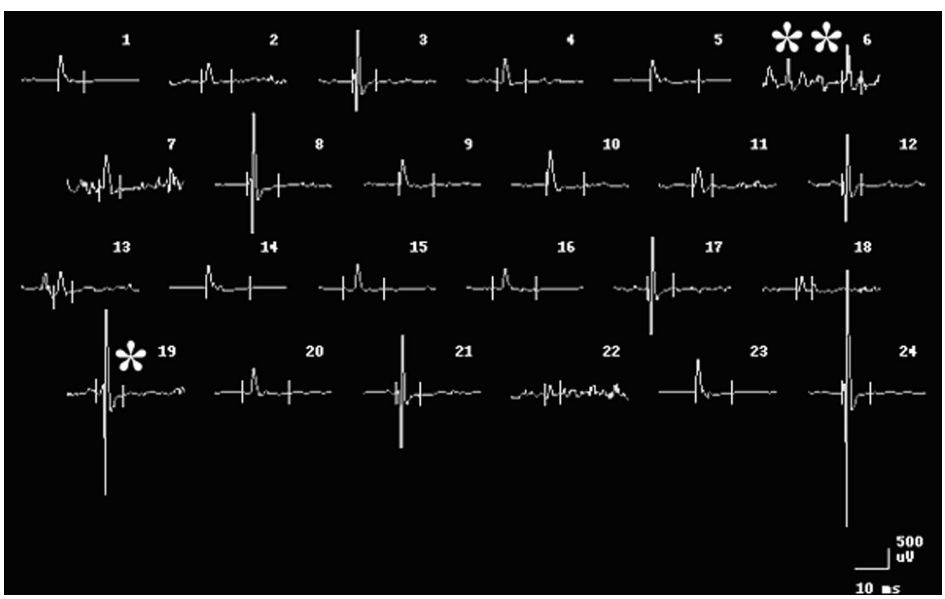


Fig. 2. Electromyography (EMG) at 15 months postlesion showed that (de novo) reinnervation had occurred. No denervation activity was noted. The motor unit potentials (MUPs) were heterogeneous in amplitude and duration. Some MUPs appeared with high amplitude and short duration and may reflect muscle fiber hypertrophy (*). Other MUPs showed polyphasia and instability, which reflects ongoing reinnervation and formation of new motor end plates (**).

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after repair after RLN resection. The results were obtained using a combination of nerve grafting microsurgery and pharmacologic treatment with nimodipine, a drug used for decades. Nimodipine may cause hypotension, but the current patient revealed no signs of hypotension, neither objectively (frequent blood pressure controls) nor subjectively, despite being an elite soccer player.

The neurophysiology correlated well with the clinical findings and laryngoscopic observations. The receptors for nimodipine are located along the whole motor neuron, and there is strong evidence that nimodipine increases the pace of regeneration by a factor of two.[8] Elegant in vivo experiments can show that pharmacologic regulation of calcium influx into the growth cone of regenerating axons may speed up or slow down the pace of regeneration.[9] Therefore, inhibitors to voltage gated calcium channels, such as nimodipine, increase the frequency of elongation movements of regenerating axons and result in a speeded up axon extension. Theoretically, nimodipine may permit regenerating axons to reach the intrinsic muscles of the larynx ahead of the previously described autonomous fibers. There may also be an effect on the misguidance profile, leaving a better somatotopical organization in the nucleus ambiguous, issues which are currently being investigated experimentally.

This case report alone can not separate the impact of the microsurgery performed from the pharmacologic treatment, but it demonstrates that functional recovery of the vocal cord is possible after RLN transection by way of direct anastomosis, and peroperative repair of the nerve should be considered in the light of other options.

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